

An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

GLENN J. WHITE^{1,2} AND ROBIN DEY³

ABSTRACT

Searches for narrowband transmitters above 30 GHz have covered one spectral line toward some tens of nearby stars. Uniformly searched, 404 ALMA execution blocks give 402 fine-channel windows of 15.3–1953 kHz toward 60 systems, covering 47.7 GHz of union bandwidth in 26 disjoint intervals; 1249 coarser windows carry no drift discrimination. Windows are filtered for an unresolved carrier drifting linearly in the stellar frame; crossings are tested against catalogued transitions at the stellar velocity, unattributed ones sought again in later blocks. For a carrier narrower than the instrumental channel, nine injections in ten return at 1.6×10^{15} W for the median system. The default drift ceiling is 12.0 nHz, 6 per cent of the Sheikh et al. (2019) range. Of the 56 crossings, 18 are attributed and 38 are not; four are in one block failing a quality criterion applied after inspection. None of the unattributed crossings recurred, including the two that lead their own control fields, though one of them is untested, while 13 attributed to circumstellar or foreground carbon monoxide do recur. That excludes a carrier holding its stellar-frame frequency to within ± 0.21 km s^{−1}, radiating through both observations, not a persistent transmitter of any kind; the remainder is consistent with chance to the control positions’ accuracy. A per-cent-level calibration residual could mimic a carrier in 9 of the windows; none holds a crossing. Line confusion costs 6.4 per cent of the coverage and 23 systems have no confirmation epoch; these, not sensitivity, limit the search.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer’s correlator setup. This paper searches those spectra for a narrowband transmitter at the star’s own position. No such signal was found.

Searches for *technosignatures* — detectable imprints of technology on the electromagnetic spectrum, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below ~ 30 GHz. Two targeted searches have reached nearby stars above that frequency. Steffes & DeBoer (1994) observed 40 solar-type stars within 23 pc at 203 GHz, and Mauersberger et al. (1996) included τ Cet and ϵ Eri among 17 targets at the same frequency. Each covered a single line, over no more than 35 MHz, with no search over drift rate and no test of persistence; Mason et al. (2025) has since searched more distant stars in Band 3. This survey covers 47.7 GHz in 26 disjoint intervals, searches a grid of drift rates rather than a fixed frequency, tests every crossing against the 512 control positions of its own field, and uses

repeat observations to test persistence.

The band has physical advantages and physical costs. Interstellar dispersion and scintillation are negligible at these frequencies, so a carrier arrives essentially undistorted, and a given aperture delivers far more gain per watt supplied. Against that, receiver system temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the same high gain narrows the beam: the 12 m benchmark of §6.3 radiates into a beam only 34 arcsec across, so the Earth must lie inside a beam that was deliberately aimed. What else makes the band hard belongs to the archive rather than to the sky, and §2 sets that out.

The signal sought is an *unresolved spectral carrier* — emission confined to a few channels, unresolved at the stellar position, and present at more than one epoch. Recurrence rather than brightness is what separates an interesting event from a detection (Sheikh et al. 2021), and the archive holds a second, separated epoch for some of these systems and not for others, so what could be searched and what could be confirmed are different extents. The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body (§4.1). “Narrowband” describes the assumed transmitter and not the resolving power of the archived channels. The targets were chosen by other programmes for other purposes, so every statistical statement here is about an archive-selected sample (§3.1).

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002)

¹ School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk

² RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK

³ VBRL Holdings, 50/5 Huay Kaew Road, Chang Phueak, Chiang Mai, Thailand

all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science (Morrison et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay & Tingay 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA’s 10 receiver bands cover 35 to 950 GHz, and one of them, Band 3, has been searched for technosignatures. The lowest, 35–50 GHz, does hold observations of 4 stars of this sample, but all from one programme, all in a continuum mode whose 31.25 MHz channels are 2048 times coarser than the narrowest channel searched here, and all proprietary until 2027: the coverage exists and cannot carry a carrier search. The band above it holds none at all. For continuity with the centimetre literature, the Arecibo planetary radar radiated 2.0×10^{13} W at 2.38 GHz (Sheikh et al. 2025); an aperture of that diameter is not realisable in the millimetre band, so the illustrative benchmark of §6.3 is instead a dish of 12 m. The limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search of comparable bandwidth and the only comparable archival one; the earlier nearby-star searches of Steffes & DeBoer (1994) and Mauersberger et al. (1996) covered a single line. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\text{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies between 25 and 277 times further away than the stars of the carrier search here, so at a fixed received flux an EIRP ratio between the two surveys would be 3–5 orders of magnitude of d^2 and almost nothing else. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min} \delta\nu$, and it has to be compared at the channel width each experiment recorded: $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$, so putting a coarse window on a fine channel arithmetically credits it with a depth it never reached. Mason et al. reach 3.8×10^{-23} – 5.8×10^{-23} W m $^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same threshold convention and on native channels throughout, the windows searched here reach 4.8×10^{-24} W m $^{-2}$ at best — a factor 10 deeper, in a channel 15.3 kHz wide — and 3.0×10^{-22} W m $^{-2}$ at the median, a factor 6 shallower in a channel 488 kHz wide. Per channel this survey is therefore the shallower of the two except in its finest windows, and its advantage lies

TABLE 1
THE COMPARISON PROGRAMMES OF FIGURE 1, EACH ON ITS OWN DETECTION THRESHOLD AND ON THE COMMON 5σ BASIS. $\text{EIRP}_{\min}$ IS THE DEEPEST LIMIT EACH PAPER REPORTS, AND THE FACTOR IS $5/(S/N)$, THE MINIMUM DETECTABLE FLUX BEING LINEAR IN THE THRESHOLD IN EVERY ONE OF THEIR OWN EXPRESSIONS FOR IT.

programme	S/N	factor	$\Delta\nu_{\text{ch}}$	$\text{EIRP}_{\min}$ (W)	at 5σ (W)
Enriquez + 17	25	0.20	2.8 Hz	10^{13}	2×10^{12}
Margot + 23	10	0.50	3.0 Hz	1.35×10^{13}	6.75×10^{12}
Price + 20	10	0.50	2.8 Hz	2×10^{12}	10^{12}
Mason + 25	5	1.00	30.5 kHz	6.91×10^{17}	6.91×10^{17}

elsewhere: 60 stellar systems within 40 pc rather than distant stars in four calibrator fields, and a frequency range reaching about nine times higher. Mason et al. also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the molecular line forest raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive with the pipeline of §4.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. For a carrier narrower than every programme’s channel that limit is a total radiated power rather than a spectral power density, so the axis is commensurable. The detection thresholds behind the published limits are not: of the 4 programmes drawn, 3 quote their limits at a threshold other than 5σ , and the minimum detectable flux is linear in that threshold in each of their own expressions for it. Table 1 gives each programme’s threshold, the factor that puts it on the common 5σ basis and the channel width it used, and the figure is drawn after that rescaling, with each published value kept beside it. The survey’s own filled symbols are bare 5σ thresholds on the same convention; the open symbols above them are EIRP_{90} , the power recovered nine times in ten through the single-epoch trigger stage, and every limit in this paper is quoted on that. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) is the other axis, the native channel width of every window searched here beside the channel of each comparison programme. What distinguishes this survey is its frequency range and the proximity of its targets, not its depth in any one channel.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

The statistical treatment this search needs is unusually detailed, and the appendices set it out in full rather than in summary.

3. THE SAMPLE

This search covers 1651 spectral windows toward 89 stars in 82 systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through

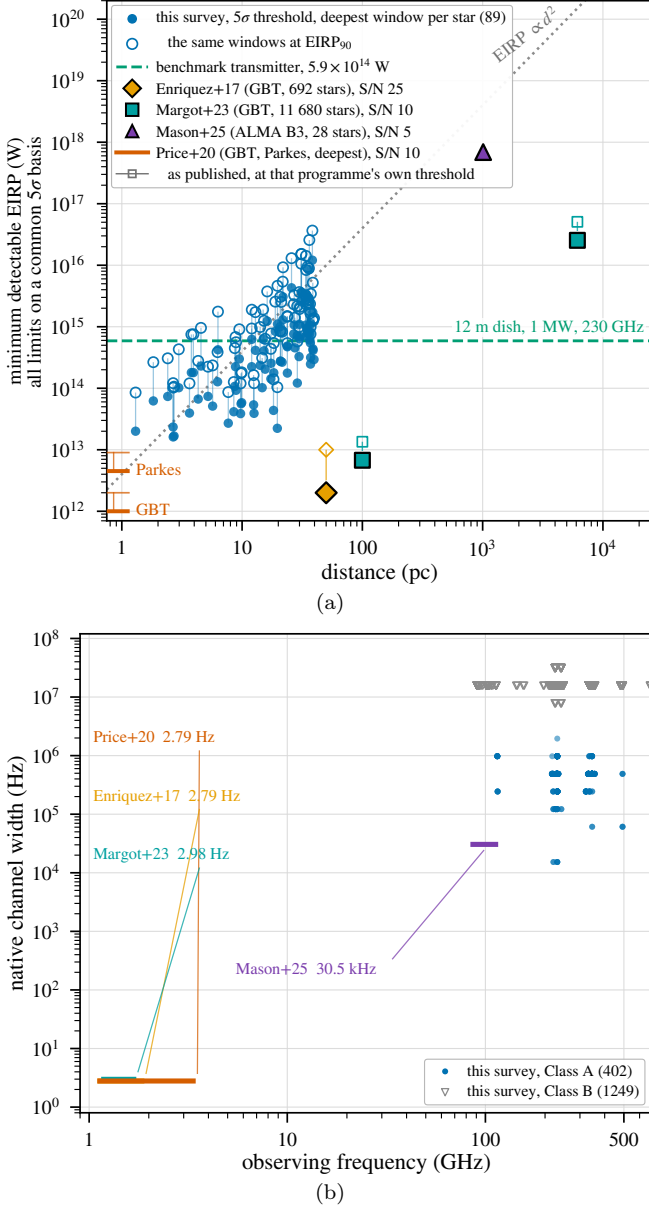


FIG. 1.— Where this survey sits among published radio technosignature searches. Every limit plotted is a total-power limit for a signal narrower than the native channel of the experiment that set it. Those channels differ by 7 orders of magnitude, from 2.79 Hz to 31.2 MHz, so the programmes are sensitive to different classes of spectral feature and a signal detectable by one of them need not be detectable by another at any power. (a) Minimum detectable EIRP against distance, all on a common 5σ basis. Blue, this survey, the deepest window per star: *filled*, that window's bare 5σ threshold, which carries no correction for the instrument's spectral response and so stands on the same footing as every other threshold in the panel; *open*, the same window at EIRP₉₀, the single-epoch trigger completeness of $\$5.1$, which does carry that response, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020). Each of these is drawn at its published limit rescaled to 5σ by the factor in Table 1, with the published value itself as the hollow marker joined above it. Grey dotted, $\text{EIRP} \propto d^2$ at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of $\$6.3$. No symbol in the panel carries a confirmation criterion. (b) Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments at the widths of Table 1.

one frozen pipeline, 280 h of telescope time and 1094 h summed over windows. That set is the *primary census*, and every limit, every crossing and every expectation here is computed over it. Three further sets of blocks enter the paper without being part of it: a hold-out reserved before any block on its own side of the rule was searched, an archival extension searched afterwards, and an out-of-sample control beyond 40 pc. A fourth, a *rank-calibration sample* drawn from the blocks already counted above, calibrates the spatial statistic. Table A1 names all five in rows that sum, and sets beside them every count of windows, crossings, systems and bandwidth this paper quotes, each with the set it counts. Wherever a number here is computed over something other than the primary census, that table's name for it is used at the point of use.

Star means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments. 402 windows toward 65 stars in 60 systems are fine-channel — *Class A* throughout — and resolve a drifting carrier; every sensitivity and every exclusion quoted here refers to them unless stated otherwise. The remaining 1249 are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess, which cannot distinguish a drifting signal from a stationary one.

The inclusion criteria were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.5. It removes 8 windows, all of them ϵ Eridani in Band 6, and no others; that band was then set aside as a whole, and Appendix A reports its limits in one place, including the 4 windows that place the star at the pointing centre and pass the floor. None of them enters a census statistic. The parallax criterion never binds: the worst fractional parallax error among the stars searched is 2.5 per cent, inside even the stricter $\sigma_\varpi/\varpi \leq 0.1$ of Table 2's reference census. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window. One target has no catalogue name — a serendipitous field star that appears throughout as ALMA J153702653–33192492, Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc with no measured systemic velocity.

Disc, planet and colour enter as descriptive metadata only, and no star was excluded by any criterion other than the three above. Three removals act on windows instead, and account for the whole difference between the 1721 rows the extraction wrote and the 1651 retained: 45 repeated rows of one window, the 12 withheld above, and 13 windows failing a *quality cut* that is physical rather than chosen. Since $\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}}$ is set by system temperature and collecting area, a window more than 100 times below the sample median in it is not measuring the sky; those removed lie 4.7×10^3 – 3.0×10^4 times below it and the nearest retained 24 times, so the threshold's value

does not matter. No integration-time cut was applied; the shortest window retained carries 21 s on source.

3.1. The archival selection

This sample was not designed: it is whatever previous ALMA programmes happened to observe, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 2 measures the bias against a volume-limited reference census of which the searched stars are a subset, classifying both columns by one rule — BP–RP colour with the absolute G magnitude, which reaches 98 per cent of the census and 98 per cent of the searched sample — so every ratio below is like against like.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 49, and the enhancement falls monotonically with distance to a factor of two under-representation beyond 30 pc. Colour comes second: A stars are over-represented by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. M dwarfs are depleted — 41 of the 89 searched stars are M dwarfs, a factor 0.7 of their share of the reference census, and only 28 of them carry the fine-channel coverage that can resolve a drifting carrier at all. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars, and none of them is an occurrence rate for the solar neighbourhood. The selection acts hardest on exactly the targets a designed search would pick: of the 15 sample systems within 15 pc with a confirmed planet, 6 carry no Class A coverage and so no drift-resolving limit.

3.2. Frequency coverage and archival scope

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band, so the endpoints are a range over which observations exist and never one that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. Fig. 2 shows the distribution; a transmitter at a frequency between the islands was never observable here, whatever its power.

The archive is not exhausted either, and the shortfall is accounted for. 177 further blocks lie inside the same query but were not processed when the census was frozen. 144 have since been calibrated and searched, 115 of them reported here as repeat epochs and as a sample the frozen pipeline had not seen; the other 29 enter no number anywhere, for the reasons Table A1 gives row by row. The largest single gap is the 18 blocks whose raw visibilities the archive holds but never pipeline-calibrated, 237 GB in all; no effort of ours would close it.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each chan-

TABLE 2

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. BOTH COLUMNS ARE THE SAME GAIA DR3 SELECTION — $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc, 17 566 OBJECTS — OF WHICH THE 89 SEARCHED STARS ARE ROWS, SO THE TWO ARE SELECTED ALIKE BY CONSTRUCTION; AND BOTH ARE CLASSIFIED BY ONE RULE, IN WHICH BP–RP COLOUR SETS THE CLASS AND THE ABSOLUTE G MAGNITUDE SEPARATES THE DEGENERATES THAT COLOUR ALONE WOULD PLACE AMONG THE A STARS. THE RULE REACHES 17 278 CENSUS OBJECTS AND 87 OF THE 89 SEARCHED STARS, THE REST CARRYING NO GAIA BP OR RP PHOTOMETRY. “RATIO” IS THE SEARCHED COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION. THE STRONGEST SELECTION EFFECT HERE IS DISTANCE, NOT COLOUR.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	12 (13.5%)	49.3
5–10 pc	240 (1.4%)	13 (14.6%)	10.7
10–20 pc	2069 (11.8%)	18 (20.2%)	1.7
20–30 pc	5391 (30.7%)	19 (21.3%)	0.7
30–40 pc	9818 (55.9%)	27 (30.3%)	0.5
Colour class ^a			
A	61 (0.4%)	5 (6.3%)	17.4
F	473 (2.8%)	8 (10.1%)	3.6
G	978 (5.8%)	15 (19.0%)	3.3
K	2221 (13.2%)	10 (12.7%)	1.0
M	13 082 (77.8%)	41 (51.9%)	0.7
Other axes			
Confirmed planet hosts ^b	26 (15.5%)	23 (25.8%)	1.7
White dwarfs ^a	463 (2.7%)	8 (9.0%)	3.4
Searched stars / systems	—	89 / 82	—

^aColour bins, not MK types. The boundaries are the colours at which the census’s own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K: BP–RP < 0.36 for A, 0.36 – 0.72 for F, 0.72 – 0.95 for G, 0.95 – 1.77 for K and > 1.77 for M. Over the 8594 census objects that also carry a Gaia temperature the two classifications agree for 98 per cent, and every disagreement is between neighbouring bins. A star is counted as degenerate when it lies more than 4 mag below the main sequence measured from the census itself: the two sequences are separated by 5 to 8 mag here, so moving the cut by a magnitude either way changes the count by under 3 per cent. The five classes are normalised over the 79 non-degenerate searched stars and the 16 815 non-degenerate census objects.

^bBoth columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscompars` table accessed 2026 October 4. The 23 searched hosts carry 50 confirmed planets, 21 of the 23 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

nel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. The de-drifted search statistic

Each window’s dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\nu, \dot{\nu}} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1/\sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν , and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation

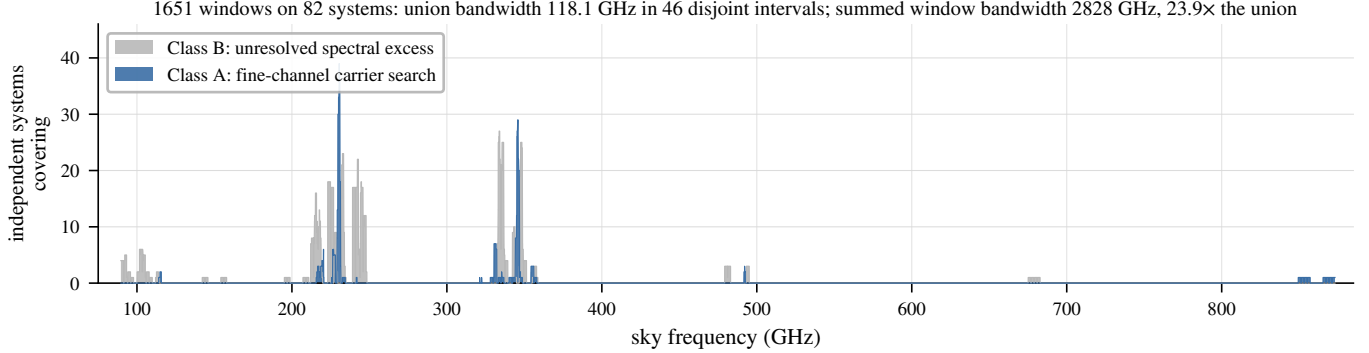


FIG. 2.— Where this survey searched. *Lower panel:* each searched window is a horizontal segment on its host system’s row, rows ordered by distance; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel:* the number of systems covering each gigahertz of sky frequency. The summed window bandwidth is 23.9× the union, so the union measures frequency coverage and the summed bandwidth search effort.

across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. T_* denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel’s effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator’s own channel shape supplies the difference (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, which is what the data contain and what a reader can recompute from the archived channel axes. Two steps need the star’s own frame and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block’s frequency axis has been corrected for the Earth’s motion at that epoch and then for the star’s systemic velocity.

Two power quantities carry the main text, both defined in Table 3: the trigger level $P_{\text{trig}} = 4\pi d^2 S \Delta\nu$ at distance d , for a carrier that makes T_* reach $T_{\text{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90} . The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline.

Three campaigns did that, each with its own denominator. The completeness itself is measured on 49 of the 402 fine-channel windows, injected directly, and on 5936 carriers over 35 windows of the coarse class. What the spatial rank costs is measured separately, on 8860 carriers over 52 of 56 attempted block-window configurations. The primary-beam correction is tested on a third set of 40 scored units, 10 of them also among the directly injected windows. Between them they span both ALMA arrays, four bands and the sample’s full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 4. Two comparisons bear on the amplitude scale and they are not the same kind of check. An independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014 , which tests the extraction and not ALMA’s flux transfer, an error in which would appear in both reductions alike; the external comparison is with the published quiescent flux density of ϵ Eridani, recovered to 1.10 ± 0.31 (Burton et al. 2022) and so a 28 per cent measurement. The absolute scale is therefore carried at ALMA’s own band-dependent accuracy specification, and the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz, accordingly carry ± 21 per cent where the sample as a whole carries ± 11 .

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot{\nu}}{\nu} = \frac{a_{\text{los}}}{c} \quad \text{and so} \quad a_{\text{los}} = \frac{c \dot{\nu}}{\nu}, \quad (2)$$

so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band. The drift and acceleration ceilings adopted for each window were computed from Equation 2 window by window, and reproduce it to 1×10^{-4} over all 402 Class A windows. The default ceiling is 12.0–13.3 Hz s^{−1} GHz^{−1}, which is 12.0–13.3 nHz in the fractional units the literature uses, and is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more; at unit margin the search would have been blind to two of the windows that produced a crossing. It is 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend, their figure being set by the rotation of small,

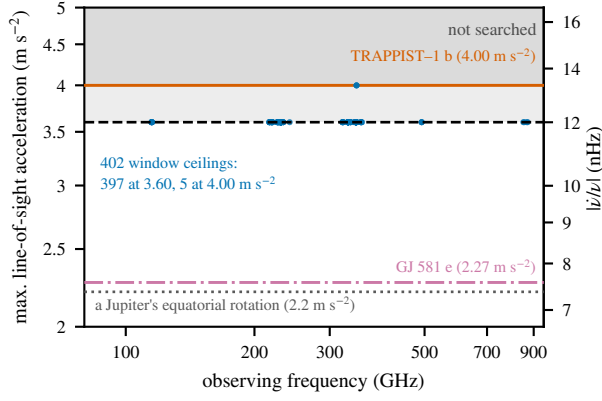


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration by Equation 2, against observing frequency; the shaded region above is unsearched. The right-hand axis is the same quantity as a fractional drift rate, $|\dot{\nu}/\nu| = a/c$, which is the unit the literature states drift reach in. The boundary drawn is the default ceiling, 3.60 m s^{-2} , equivalently 12.0 nHz , which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST-1 windows. Horizontal lines are reference accelerations: the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}), and the equatorial rotation of a Jupiter, 2.2 m s^{-2} or 7.4 nHz . Earth’s own rotation and orbit, 0.034 and 0.0059 m s^{-2} , lie below the foot of the axis. **Transmitters whose combined orbital and rotational acceleration exceeds the ceiling are not constrained by this search, and the ceiling is a budget they share:** a transmitter on the equator of a rotating giant spends 61 per cent of it on the planet’s spin alone. Nor are unbound transmitters, or any platform accelerating for its own reasons.

fast solar-system bodies rather than by a planet’s own orbital acceleration, and the restriction here is deliberate. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Three limits follow: a deliberately chirped carrier is not distinguished from an accelerating platform; a transmitter need not be bound to a known planet, so drifts faster than the ceiling are unconstrained; and orbital and rotational acceleration add, so a transmitter on a rapidly rotating body reaches the ceiling on its spin alone. For the most demanding known case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling and 1 per cent above the widened grid used on that star’s own windows (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 3). Channelisation in the sample is bimodal with an empty gap between the modes, so the split is unambiguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta \nu_{\text{ch}} \gtrsim 1$. Each window’s own η_{drift} , trial-drift count and acceleration ceiling follow from its channel width, its track length and Equation 2, so the sample may be reclassified on that criterion instead; the two do not partition the survey identically. The two classes are physically different experiments, results are given per class, and they are combined only where that is appropriate.

4.2. From a crossing to a confirmed signal

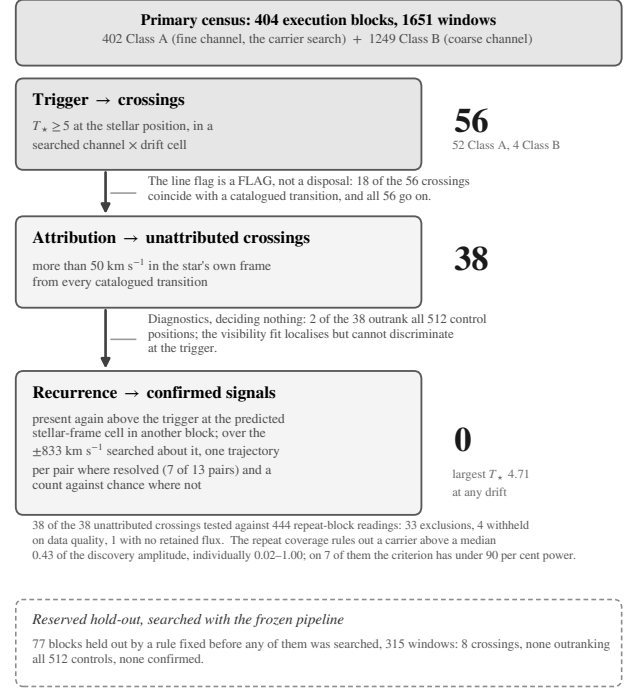


FIG. 4.— The detection logic, with the count at each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and the quality-failing block that no expectation is computed over, so the figure, Table D1 and the text cannot disagree. The counts drawn are the primary crossings, one to a window; the 21 secondary crossings of §5.3 enter the same three levels and three reach the second, none the third.

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 4). Each is set out below as the quantity it measures, the number that quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

- (i) *Trigger → a crossing*. The quantity is T_* , the derived statistic of Eq. 1 read at the stellar position; the criterion is $T_* \geq T_{\text{trig}} = 5$ in at least one channel $\times$ drift cell; and both the statistic and the level were fixed before the archival search began. **T_{trig} is a trigger threshold and not a detection significance:** T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. Every cell that reaches the level is recorded, not only the largest in each window, because a carrier is masked as effectively by a noise excursion as by a line. A crossing establishes only that it goes on

to the next two tests.

- (ii) *Line attribution* $\rightarrow$ *an unattributed crossing*. The quantity is the crossing’s velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star’s own rest frame; the criterion is $\pm 50 \text{ km s}^{-1}$. The statistic and the trigger were fixed before the archival search began; the line catalogue is a single database query made afterwards, on 2026 October 6, so the dispositions rest on the physical evidence set out beside each one and not on the order in which the list was assembled. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed* where the species is one that such systems are known to carry: a coincidence with a catalogued transition makes an astrophysical origin far more likely than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. A further three crossings coincide with a species no debris disc has been shown to carry; their offsets are reported and they are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**, wherever a later block covers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window’s price is that the survey constrains nothing inside it (Appendix C).
- (iii) *Independent recurrence* $\rightarrow$ *a confirmed signal*. The quantity is $T_\star$ at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive’s uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks $T_\star$ against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block’s own synthesised and primary beams so that no control sits on the star; its lowest value, $1/(512 + 1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing’s emission is at the stellar coordinates rather

than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank’s own null is called one.

One execution block is not a null, and a uniform rule says so. The comparison with the control positions needs each block’s ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 404 primary-census blocks, 1 fails — A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst is 1.12. The Class A results are identical with it and without it, the same 402 windows and 52 crossings either way, because its 4 windows are all coarse-channel; what it does hold is 4 of the 56 crossings, the survey’s entire Class B crossing population (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window’s own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax, whose omission is corrected window by window in §5.1; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the primary census. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a recovery completeness of one epoch, and not a threshold. Artificial carriers were injected at a random sub-channel phase and a random drift from each window’s own grid and recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP₉₀, the only sensitivity quoted here, measured through the single-epoch trigger

TABLE 3

THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER. THE DISTINCTION THE FIRST THREE ROWS DRAW IS THE ONE THE LIMITS OF §5.1 TURN ON: A CROSSING IS A SINGLE-EPOCH EVENT, AND A CONFIRMED SIGNAL NEEDS A SECOND AND SEPARATED EPOCH THAT NO SENSITIVITY SUPPLIES.

Term	Definition
Crossing	A channel $\times$ drift cell at the stellar position reaching $T_* \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its <i>primary</i> crossing and any other a <i>secondary</i> one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than 50 km s^{-1} in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	An <i>unattributed</i> crossing recovered at the predicted stellar-frame frequency and drift in an independently separated block. The criterion is applied to every crossing, and a crossing coincident with a catalogued transition that satisfies it is a recurring molecular line: 13 of them are, which is this stage's positive control (§5.6). The reading is taken at one channel at one drift, so it holds only for a carrier whose stellar-frame frequency moved by between ± 0.08 and $\pm 0.63 \text{ km s}^{-1}$ at most between the epochs compared; over the $\pm 833 \text{ km s}^{-1}$ also searched about that cell, a pair of crossings is tested against a single bounded trajectory where the drift grid resolves the interval between them (7 of 13 pairs, Table 5). It can be applied only where the archive observed the same frequency again, so for the 23 Class A systems where it did not, no sensitivity makes a confirmed signal possible.
P_{trig}	The power of a carrier that makes T_* reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP ₉₀	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 353 an estimate from the class-average calibration.
Class A	The primary experiment: 402 fine-channel windows, median 488.3 kHz, toward 60 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.
Class B	The secondary experiment: 1249 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

and the localisation at the stellar position inside one execution block. **It is therefore the power at which a carrier would have produced a crossing and not the power at which one would have been confirmed**, because confirmation needs a second and separated epoch, which the archive supplies for some systems and not others whatever their depth (§5.6.1). The 512-position spatial rank decides nothing and is not charged against it either: charging it would cost a further factor 1.31 and in 3 bright-ring windows could not be met at any injected power (Fig. 5).

Measured that way the completeness is one number for the class. Nine carriers in ten are recovered at $3.03 P_{\text{trig}}$; the sampling error of that pooled value is ± 2 per cent, which is a statement about the class average and about no individual window. It is a total power set against a nominal threshold: P_{trig} is the bare 5σ of the window itself, and the loss a channel-confined carrier suffers to the correlator's spectral response is carried by the injected tones, so it sits inside the measured factor — $\times 2.29$ of it — and is not applied to the result as well (Appendix A). Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, and the value does not depend on the brightness of its control ring: the 5 windows whose ring carries resolved emission come back at $3.24 P_{\text{trig}}$ against 2.94 for the rest (Fig. 5). Class B carries its own factor $4.17 P_{\text{trig}}$, measured the same way from 5936 carriers injected into 35 coarse windows of 14 execution blocks; it is what sets the limits for Barnard's Star, Wolf 359 and Proxima.

The resulting limits. The median Class A window reaches $2.9 \times 10^{15} \text{ W}$ of equivalent isotropic radiated power and the median system, on its best window, $1.6 \times 10^{15} \text{ W}$, with a spread across systems of 8.7×10^{13} – $3.7 \times 10^{16} \text{ W}$ (Fig. 6a). The round 10^{15} W level is reached for 24 of them. The calibration of the factor is good to $\times 0.89$ – $\times 1.11$ (Table 4); that is the interval on the median and on the counts, and not on any one window's limit. One bias is not inside it, runs one way only, and is *not* included in the values quoted here: an injected carrier is added to already calibrated visibilities and so suffers no atmospheric decorrelation, whereas a real carrier does. Measured block by block from the phase rms the observatory records on each block's phase calibrator (Cortes et al. 2026, §10.4.9), through the standard phase-scatter relation (Richards et al. 2022, Eq. 3), it leaves the median limit optimistic by 1 per cent and the median window above 400 GHz by 8 per cent, so the limits at the top of the frequency range are the least secure quoted here. **The 125 blocks whose delivery exposes no such record are not a random subset of the 404:** their longest baseline has a median of 279 m against 46 for the 279 that do, and it is the long baselines that lose coherence. Matching each of their windows to the measured blocks within a factor of two in baseline length puts the correction at 1.9 per cent in the median and 21 at the ninetieth percentile, so their limits are conditional on a calibration performance the archive does not report.

Two further losses act window by window, and both are divided out of the limits above rather than quoted beside them (Appendix A). The search's phase centre omits the annual parallax, which displaces the star by at most 0.67 of a synthesised beam and makes the median Class A limit shallower by 1.5 per cent, while Wolf 28

in A002_X9f2ff8_X25b0 has no astrometric keys and is left uncorrected. That correction is concentrated rather than general, which is why the median limit moves far more than the median window's own factor (Table 4): 216 of the 402 windows need none at all, while 14 lose more than a tenth and lie around the middle of the distribution of limits. The second loss the injections do not reproduce at all, each carrier being laid down at one frequency per integration while a real one sweeps within it. **It costs nothing in the median window and up to $\times 1.53$ in the 13 finest-channel ones, which are also the deepest:** the best Class A window moves from 6.8×10^{13} to 8.7×10^{13} W, and HIP 10679 drops out of the 25 systems that would otherwise reach 10^{15} W.

Carriers were injected directly into 49 of the 402 Class A windows, 12 per cent, and the result transferred to the remaining 353, which is the dominant caveat on all of the above. The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance and without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block's calibrated visibilities are what an injection campaign has to rebuild, so band and array were fixed by the draw and the conditions that vary within a block were compared with the population afterwards. **That comparison covers 6 properties** — channel width, integration time, noise, controlling brightness, the channels a carrier at the drift ceiling crosses in one track, and the intra-integration retention — and on every one the injected windows span the central eight-tenths of the population, five agreeing in the median to 6.3 per cent. The sixth, the noise level, does not: the injected windows are $\times 1.7$ noisier, which follows from drawing whole blocks, and it is also the property the measurement is built to be insensitive to, each ladder being in units of its own window's threshold. That is measured rather than assumed: splitting the injected set at the median of each property in turn moves the ninety per cent point by at most $0.27 P_{\text{trig}}$. The draw controlled the array, but the achieved mix still departs from the population by 22 per cent of the sample; re-weighting the pooled curve by the survey's own composition gives $2.98 P_{\text{trig}}$ against the adopted 3.03.

A window that was not injected therefore carries an estimated EIRP₉₀, and across the injected windows the ninety per cent point ranges from $\times 0.80$ to $\times 1.58$ of the adopted factor: that dispersion, not the calibration interval above, is what one uninjected window's own limit is uncertain by. Because a system's limit is its best window, a minimum over several such estimates could favour windows whose own factor falls below the class average; drawing each window's factor from the 49 measured ones instead, over 4000 trials, moves the per-system median deeper by 2.2 per cent, holds the count at 10^{15} W to 23–24 in nine trials in ten and spreads one system's own limit by -17 to $+9$ per cent, so the estimate is mildly conservative. But **the best window was itself injected into for only 6 of the 60 systems:** for the other 54 the headline limit is an estimate.

A carrier that is not always radiating is penalised in proportion. The de-drifted stack is coherent across a whole execution block, so over 3888 trials at duty cycles down to 10 per cent the recovered amplitude followed the

TABLE 4
WHAT SETS EIRP₉₀, AND IN WHICH DIRECTION. THE UNCERTAINTY ON THE CALIBRATION IS AN UNCERTAINTY ON THE CLASS-AVERAGE FACTOR, AND SO ON ANY STATISTIC FORMED OVER MANY WINDOWS; ITS FLUX TERM IS BAND-DEPENDENT, SO THE SUBMILLIMETRE WINDOWS CARRY THE LARGER COMBINED FIGURE. THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW'S OWN LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR, AND THE DECORRELATION TERM IS MEASURED IN 279 OF THE 404 BLOCKS AND NOT IMPUTED TO THE REST.

	value	what it does to the limit
Class-average completeness	$\times 3.03$	nine carriers in ten recovered at this multiple of a window's own trigger power. <i>Measured</i> in the 49 Class A windows injected directly and <i>estimated</i> , at this factor, in the other 353
Uncertainty on that calibration (per cent)	± 2	combined in quadrature sampling error of the pooled value over the 49 injected windows
Absolute flux scale	± 10.2	the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): ± 5 in Bands 1–5; ± 10 in Bands 6 and 7; ± 20 in Bands 8–10, weighted over this sample's bands
Distance and visibility scale	± 4.5	the Gaia parallaxes, which enter as d^2 , and the block-to-block visibility scale
Combined	$-11/+11$	$\times 0.89$ – $\times 1.11$ on the factor, and so on the median limit and on the system counts; ± 21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Spread between windows. An observed dispersion, not an error bar.		
Window to window	$\times 0.80$ – $\times 1.58$	the 49 injected windows resolve 2.41–4.80 P_{trig} individually. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 54 of the 60 systems
Applied. Folded into every limit quoted, and so not combined with the above.		
Intra-integration smearing	$\times 1.00$ – $\times 1.53$	a carrier sweeps through part of a channel within one integration and the injected ones do not: η_{smear} at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1046 windows) and from the longest delivered anywhere where it is not (605)
Annual parallax	$\times 1.00$ – $\times 2.06$	the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	$\times 2.29$	the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of $\times 1.325$ over the nominal trigger
its residual	$+8$	the injected profile falls short of the exact response: conservative
Decorrelation	$+0/+1$	measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does: $+8$ above 400 GHz, and more than $+20$ in 5 per cent of windows

duty cycle to 1 per cent: a transmitter radiating a tenth of the time needs $10\times$ the power quoted above. Those trials carry no drift, so the joint drifting and intermittent case is not measured here.

5.2. The nearest stars: Barnard's Star and Wolf 359

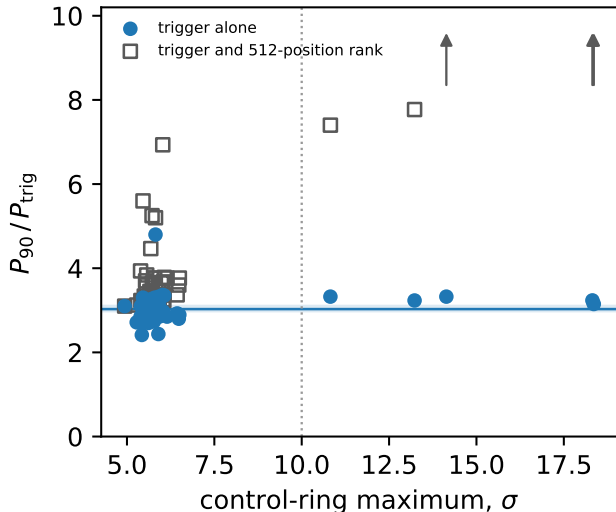


FIG. 5.— Why one completeness factor serves the whole class. Each of the 49 of the 402 Class A windows injected directly is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here — the trigger and localisation at the star — with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission rather than noise.

Barnard’s Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search*: the archive holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the EIRP₉₀ convention of §5.1 Barnard’s Star reaches 2.7×10^{14} – 4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14} – 4.5×10^{14} W over 12.

5.3. Threshold crossings and how each was disposed of

The result is a chain of three tests, and it is read in one direction. Of the windows the search covers — the 1651 of the primary census and the archival-tail blocks beside them — 56 hold a cell reaching $T \geq 5$ at the stellar position. 4 lie in the one execution block that fails the quality criterion of §5.3.2, fixed before it was applied, so the population that characterises the search is the 52 in quality-passing data and those 4 are reported beside it. The molecular-line mask, $\pm 50 \text{ km s}^{-1}$ wide in each star’s own rest frame, places 21 of the 52 within that distance of a catalogued transition; three of those coincidences are with a species no debris disc has been shown to carry and are reported rather than acted on (Appendix C), leaving 18 attributed and 34 unattributed. The attributions are kinematic and not an accident of bandwidth: in the delivered frame, 11 of the 51 crossings with a measured peak frequency fall within $\pm 50 \text{ km s}^{-1}$ of a transition these setups were tuned to, against 4.7 expected were each peak free to land anywhere on its own window’s grid (Appendix E). Of the unattributed, 2 outrank all 512 control positions of their own window, and none was

found again in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 4 counts what survives each step and Tables D1 and D2 list every crossing.

Anything bright in a window hides a weaker carrier behind it, so the weaker cells were sought too. The de-drifted statistic is retained in every channel, so behind a window’s tabulated crossing — its largest cell — weaker above-trigger cells can be read as well, neighbouring channels counting as one. Re-read that way, the 399 of the 402 Class A windows that retain such a profile — the other 3 retain none — hold 49 maxima at or above the trigger and twenty-one weaker cells besides, in 9 windows and as many as 8 in one. 18 are the resolved carbon monoxide of a disc spread over channels; the three unattributed ones, toward HD 14055, 61 Vir and β Pic, each have 3–16 covering blocks and none reaches the trigger in any of them, the deepest repeat being $T_{\star} = 4.4$. Applied to the window maxima the rule returns 50 of the 56 tabulated crossings to 0.002 of a channel; the 6 it misses are the 4 of the failing block, whose repaired profile was not retained, and crossings whose own window retains no usable profile. Cells are counted in the controls too, and outside the mask the star holds 1.05–1.15 times as many as a control position at every level from $T = 3$ to the trigger, with no step where a carrier search would put one; the expectation that follows is §5.3.1’s.

5.3.1. Most of the crossings are expected by chance

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over the 402 Class A windows of the census and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. A window’s own 512 control positions measure how often a position in that field reaches the trigger, and the window’s own channel grid measures how often such a position would fall inside an attribution window, which takes 20 per cent of the total. The control positions differ from the star in one respect that bears on this: they carry no continuum, so a calibration error that varied with frequency would raise the star and not them. Measured on the continuum these stars are found to have, the windows where such an error could reach the trigger hold no crossing and bound it below 6.6 per cent (Appendix B). Chance predicts 28.9 unattributed Class A crossings, with 12–48 allowed by resampling the execution blocks the windows come from, and 32 are found. Counting cells rather than windows on both sides moves the two together: the same 3192 control positions predict 28.1 cells, 2.2 of them beside a window’s largest, against 34 and 3 found and 13–48 allowed, because a control position holds a second above-trigger cell about as often as a star does (§5.3). A further two unattributed crossings, both toward HD 14055, lie in archival-tail blocks the catalogue does not tabulate window by window, and enter neither side.

The expectation is a sum of rates measured at control positions, and the reserved blocks measure how far a star departs from them: carrying that difference through moves it to 30.8, and the count agrees with chance either

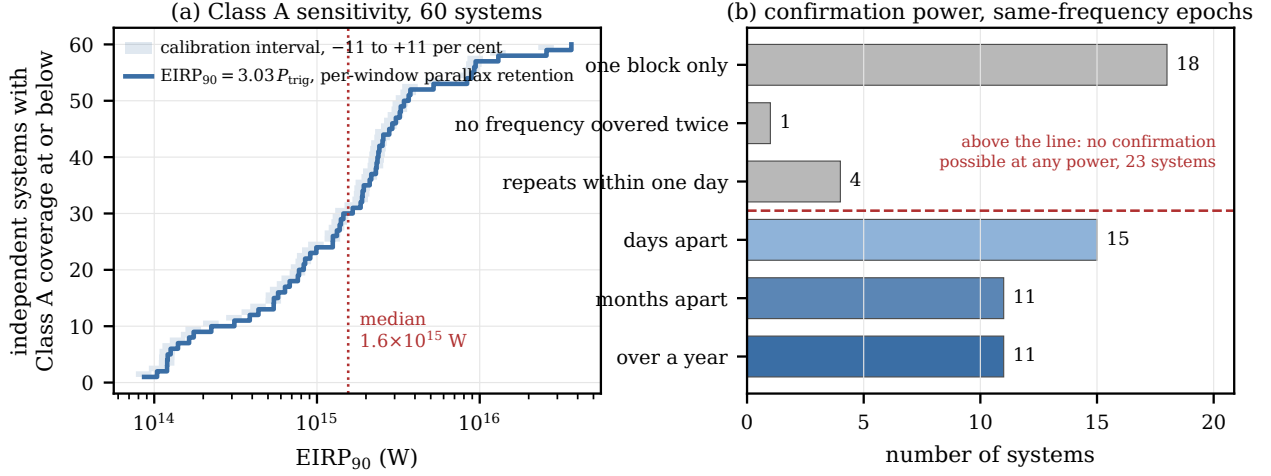


FIG. 6.— **The two completenesses of the carrier search, over the same 60 systems.** (a) Single-epoch trigger completeness: the cumulative number of systems whose best window reaches a given EIRP₉₀ (Table 3), evaluated as $3.03 P_{\text{trig}}$ with each window’s own annual-parallax retention applied; the shaded band is the calibration interval of Table 4. (b) Confirmation completeness, which no amount of power buys: systems grouped by the longest separation between two execution blocks covering a common frequency, more than 1 day counting as independent. **Left of the dashed line, 23 systems have no such pair, and for them confirmation was not possible at any sensitivity in panel (a): what panel (a) gives them is a crossing limit, not a detection limit.**

way. The width of the range allowed is what the comparison can say and no more — an added population of as many as 19 crossings would sit inside it unremarked. What tests each crossing individually is the recurrence test of §5.6.

5.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), all in one ACA execution block, A002_Xc7fa6f_X28a8, whose whole field is above threshold: 512 of 512 control positions exceed $T = 5$, where the median of a window’s own control ensemble is 2.75 across the survey. Its single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal — so a slow drift common to every channel and position inflates the statistic across the field, while the 24 sibling executions of the same field, array and setup reach only $T_* = 4.26$ and produce no crossing. *What is inflated is the field and not the star:* the strongest of the four reaches $T_* = 73.4$ where its own controls peak at 74.7, and 1–421 of the 512 stand above the star. The criterion fails this block and no other (Appendix D.2), so its 4 crossings are reported apart from the 52; Appendix E carries what the block might instead be.

5.3.3. Stellar flares are excluded by a broadband prediction

The sample contains known flare stars. But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4717 σ , against an observed continuum statistic of median -0.29 with $|Z| < 3$ in 45 of them. One block settles it: AU Mic A002_Xa42f75_X319 holds both a crossing ($T_* = 5.27$) and the millimetre flare of MacGregor et al. (2020), and *that flare contributes 0.17 σ to the single-channel statistic* — raising the continuum at $Z = 10.3$ while the crossing’s own de-drifted track reaches $Z_{\text{max}} = 2.1$.

5.4. β Pictoris: an astrophysical positive control, and the obstacle it identifies

A search that finds nothing must show that it could have found something, and the survey’s strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 10 independent execution blocks spanning 8.4 yr, and in 9 of those the star also outranks every control: 5 Band 3 windows at T_* up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block’s own epoch, then to the star’s rest frame — all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block’s own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in an independent execution block, the screened Band 6 window returning $T_* = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject. The rank is therefore a test of compactness and not of authenticity, and on a real extended source it errs conservatively — a stronger statement than a clean pos-

itive control, because it bounds the diagnostic’s reach as well as demonstrating it (Fig. B1).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block’s $^{13}\text{CO}(2\rightarrow1)$ window is bright at the ring and faint at the star. What identifies the extended emission is therefore the *level* of the control ring and not the star’s rank against it — the star here outranks every control — which is why the window-level flag of Appendix D.2 is defined on the ring’s median.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 18 of the 56 crossings land on a catalogued transition. The mask that removes them costs 6.4 per cent of the Class A union bandwidth over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a number on it.

5.5. The two crossings that lead their own control fields

Frequencies in this section and the next are in the star’s own rest frame, which is the frame the recurrence test registers in; the delivered ones are in Table 6 and in the ledger.

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.3232 GHz and CP–72 2713 at 344.2965 GHz, both in Band 7. Both are tabulated in Table 6, the first is drawn in Fig. 7, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4). CP–72 2713’s crossing lies -12 km s^{-1} from $\text{SO}(8_8-7_7)$, a coincidence reported as a note rather than as a disposition (Appendix C). Both events go on to the recurrence test. Each was also re-fitted in the visibility domain, as 51 of the 56 crossings were, by phase rotating to the star and following the recovered drift; 41 of those fits are consistent with a point source at the star, and injected sources in the bin these two occupy return $\text{Re}/\sigma = 6.58 \pm 1.97$, so the fit excludes none. Its value is that a pedestal fed through a sidelobe from elsewhere in the field, which a dirty map at the star cannot tell from a source at the phase centre, is not a point source there. Of the 512 control positions the rank screen uses, 8 retain a dynamic spectrum and 12 are reachable in the visibilities.

Neither crossing is present when the same stellar-frame frequency is observed again. The reading that assumes least is the nearest covering block alone, taken 1.97 and 2.05 hours from each discovery: a carrier persisting at the discovery amplitude would have given $T_{\text{pers}} = 5.28$ there against -0.01 measured for 61 Vir and 6.21 against -0.57 for CP–72 2713 — whose nearest block is also its deepest — ruling out anything above 0.95 and 0.71 of that amplitude. Over intervals of hours a transmitter orbiting at 1 au stays inside the ± 0.211 and $\pm 0.213\text{ km s}^{-1}$ half-channel those readings cover, so both survive an orbit of that period. Combining all 3 and 5 covering blocks

deepens the limits to 0.56 and 0.66 of that amplitude, but requires the carrier to hold one channel across baselines of 3 and 2 days, over which the same orbit would move it 1.53 and 1.08 km s^{-1} : those readings constrain only a carrier stable to half a channel over the whole baseline. At the faintest amplitude consistent with each crossing having been found at all, the criterion would have recovered a persisting carrier with probability 1.00 and 0.66: for CP–72 2713 the silence is not an exclusion, and is not read as one. Appendix B follows CP–72 2713’s crossing through every step of the chain.

5.6. Recurrence, applied to every unattributed crossing

A carrier that stays inside the repeat window and radiates through both observations must be there when the same frequency is observed again. That is the confirmation criterion, applied in the star’s own frame: the predicted channel follows from the Earth’s motion at the repeat epoch and the star’s systemic velocity, the discovery drift is carried to it, and the statistic is read there, maximised over nothing. The frame matters, because for η Crv it moves the predicted channel by -80.2 channels. A *confirmed signal* is an *unattributed* crossing that satisfies the criterion (Table 3); a line-coincident crossing that satisfies it is a recurring molecular line, and is reported as one.

None of the unattributed crossings recurred, while 13 crossings attributed to circumstellar or foreground carbon monoxide do recur, as expected. Those 13 are the positive control of this stage and they test it on both of the ways real emission arises here: 10 are the resolved circumstellar disc of β Pic and 3 are foreground gas toward HD 285968, whose sightline carries catalogued cloud emission at the crossing’s own velocity in the local standard of rest (Appendix C). No exclusion is formed for them, because what one would exclude has just been observed. Close to the trigger the control is thinner than that count suggests: every circumstellar recovery has $T_\star \geq 8.0$, and the 2 attributed crossings that are not recovered are the two faintest tested, both at $T_\star = 5.41$, so there the stage is controlled by HD 285968 alone — by foreground gas rather than by a point source. All 38 unattributed crossings were tested, over 444 readings in 151 execution blocks, and at the predicted cell the largest statistic over all of them is $+2.00$ at the carried drift and 4.71 over every drift trial, against a trigger of 5.

Every exclusion is conditional on a requirement of frequency stability, and it is a narrow one. The reading is taken at one cell at one drift, so it constrains only a carrier whose stellar-frame frequency moved by less than half that cell between the epochs compared: 488 kHz, which is ± 0.08 to $\pm 0.63\text{ km s}^{-1}$ across the 33 crossings that carry an exclusion, median ± 0.21 . A transmitter in a circular orbit at 1 au accelerates along the line of sight by up to 5.9 mm s^{-2} , which keeps it inside that interval for 14 of the 33 over the interval to the nearest covering block and for 1 of them over the longest baseline; at the 3.6 ms^{-2} the drift grid reaches, none of the exclusions survive. The search accommodates planetary acceleration and this test does not, so the two bear on different classes of transmitter: non-recurrence excludes a carrier that holds its frequency within those limits and radiates through both observations, and says nothing at all about one whose orbital motion carries it out of the cell. That

is why the whole window is searched at every drift of the grid at each repeat epoch, which spans that excursion for 30 of the 32 crossings another block covers.

The amplitude each exclusion reaches is the fraction of the discovery amplitude still excluded at the trigger, $(T_{\text{rep}} + 5)/T_{\text{pers}}$. Of the 38 crossings tested, 33 carry a quoted exclusion; 4 take their discovery amplitude from the block that fails the quality criterion of §5.3.2 and 1 retains no discovery spectrum, so none can be formed for those. Over the 33 the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude and the deepest at 0.017 of it, but the individual limits run to 1.00, and for 3 of them — β Pic at 241.77 and 241.86 GHz and HD 110411 at 219.44 GHz — nothing below 0.84 is ruled out. Combining every covering block buys most of that, and requires the carrier to return to the same cell at the start of each: the nearest block alone rules out a median 0.98 of the discovery amplitude, and 15 of the 33 rule out nothing. The chance that a carrier would have satisfied the criterion is evaluated at the trigger amplitude $5S/T_*$ rather than at the discovery amplitude, which is selected as the largest of many cells at a statistic of about five and is biased high (Appendix B): it exceeds 90 per cent for 26 of the 33 and for the remaining 7 it is β Pic at 241.86 GHz (0.38), ALMA J153702653–33192492 at 347.19 GHz (0.50), β Pic at 241.77 GHz (0.61), CP–72 2713 at 344.30 GHz (0.66), CD–57 1054 at 346.53 GHz (0.82), AU Mic at 230.67 GHz (0.85) and HD 110411 at 219.44 GHz (0.88). Those 7 are untested rather than excluded. These are conditional detection probabilities under the persistent-carrier model adopted here, not probabilities that a candidate was noise: a crossing that does not recur has not been shown to be anything else.

The 28 repeat windows that do hold something above the trigger are themselves crossings, so a carrier present again would appear in its repeat block as one; their number agrees with the 28 chance predicts in trials of that size, but each pair must still be asked whether one transmitter could have produced both members. Table 5 asks it of all 13 pairs, across 5 stars, whose two crossings lie within 833 km s^{-1} of each other in one star’s frame. Both fitted drifts are carried into that frame first, by removing the observatory’s own line-of-sight acceleration: at most 40 Hz s^{-1} here, but diurnal, so extrapolating it across days would add an excursion that is not there.

Two criteria follow, and they partition the pairs differently. Where the drift grid resolves the interval — 7 of the 13 pairs — the elapsed time and the two drifts fix the offset a carrier at constant line-of-sight acceleration would show, and all 7 fail that comparison, by 3.6 standard deviations in the mildest case. The weaker demand is any trajectory whose drift stays inside the searched grid and whose curvature is bounded by the same acceleration ceiling, and it is the one each verdict rests on: it excludes 3 pairs — 1 needing a mean drift the search never covered and 2 needing an acceleration reversal 1.3 to 2.4 times faster than the 21 hour quarter period the ceiling allows — and leaves 10 not separable, among them 4 of the 7 that the constant-acceleration comparison resolves. β Pictoris is judged by the same criterion. Its two crossings sit at 241.7739 and 241.8594 GHz in blocks 24.4 hours apart, 106 km s^{-1} apart, and both fitted drifts are negative, -1611 and -679 Hz s^{-1} , so a single carrier

would have had to rise in frequency between the epochs; that needs a mean line-of-sight acceleration of 1.21 m s^{-2} , inside the 3.6 m s^{-2} ceiling, so the sign settles nothing on its own. The reversal does: the pair requires one in 16.5 hours, 1.3 times faster than the ceiling permits, which is the narrowest margin of the 3 exclusions and the only one inside a factor of two. The same test accepts the one pair that is a single emitter, the CO(2–1) line of HD 285968 recovered at one stellar-frame frequency 5.0 days apart, 0.6 standard deviations from the prediction. For the 10 pairs it cannot separate, 2 days to nearly two years apart, nothing measurable tells one carrier from two, and only the agreement of their number with chance bears on them.

The 2 attributed crossings of the 15 tested that are not recovered both remain consistent with their attribution. HR 1010’s crossing at 230.5287 GHz is -12.0 km s^{-1} from CO(2–1) and is bounded at $T_{\text{rep}} < 4.6$ against the $T_{\text{pers}} = 8.4$ real carbon monoxide at the discovery flux would have given, a disagreement of only 2.1σ once that window’s noise is measured over the 47 minutes it integrates, 11.5 mJy, where its sibling blocks also lie. ALMA J153702653–33192492 is a serendipitous field star with no measured systemic velocity, so its frame is the barycentric one; its crossing at 230.5392 GHz is $+1.6 \text{ km s}^{-1}$ from the same transition and the bound there is $T_{\text{rep}} < 4.2$ against $T_{\text{pers}} = 6.2$, or 1.4σ . Its attribution rests on a velocity rather than on this test: the sightline carries catalogued foreground emission at $+4.9 \text{ km s}^{-1}$ in the local standard of rest, 0.5 km s^{-1} from the crossing’s own velocity in that frame (Appendix C). What its repeat blocks show is where that gas lies rather than whether it can be recovered. All 6 were taken in the same 7 m configuration with synthesised beams of 4.8 to 5.3 arcsec, so an extended cloud is recovered alike in each, and in every one the control ring reaches 10.9 to 13.8 while the phase centre reads 4.2 to 4.8 with 193 to 496 of the 512 control positions above it: the gas is bright and it is not at the phase centre. Toward HD 285968 it is, and over the 4 covering blocks the ring reaches only 6.4 to 8.1 against 4.8 to 6.0 at the star. Whether a foreground line recurs at the phase centre is set by where the cloud is bright, which the control ring measures in the same map.

5.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements. This experiment searched 60 systems and 402 Class A windows over 47.7 GHz of union bandwidth, and could have confirmed a signal in 37 of those systems — those observed again more than a day later at a frequency the first block covered — and in 11 of them more than a year later (Fig. 8). Table A2 prints it beside the larger count that ignores frequency, and retuning is the difference.

An independent execution block is also not an independent epoch. The baselines run from 0.46 hours to 1.8 yr, and for 21 of the 37 crossings carrying an amplitude the nearest covering block falls within a day. Over months and years these tests bound no duty cycle, and for the other 23 systems they bound nothing at all.

5.6.2. The reserved hold-out and the control beyond 40 pc

TABLE 5

CAN TWO CROSSINGS BE ONE CARRIER? EVERY PAIR OF CROSSINGS TOWARD ONE STAR, IN TWO EXECUTION BLOCKS, WITHIN 833 km s^{-1} OF EACH OTHER IN THE STAR'S FRAME; STARS ARE IDENTIFIED BY POSITION, SO THE TWO ARCHIVAL-TAIL CROSSINGS OF HD 14055 PAIR WITH ITS CATALOGUED ONES. $\Delta\nu$ IS THE STELLAR-FRAME OFFSET AND $\dot{\nu}_1$, $\dot{\nu}_2$ THE DRIFTS FITTED INDEPENDENTLY IN THE TWO BLOCKS AND CARRIED INTO THAT FRAME. A TRANSMITTER AT CONSTANT LINE-OF-SIGHT ACCELERATION WOULD PUT THE PAIR AT $\Delta\nu = \frac{1}{2}(\dot{\nu}_1 + \dot{\nu}_2)\Delta t$, AND *miss* IS HOW FAR IT IS FROM THAT IN UNITS OF THE DRIFT GRID'S RESOLUTION CARRIED OVER Δt ; A DASH MEANS THAT RESOLUTION EXCEEDS THE WINDOW SEARCHED, SO THE COMPARISON HAS NO POWER. THE LAST COLUMN IS THE VERDICT UNDER THE WEAKER CRITERION OF A BOUNDED TRAJECTORY INSIDE THE SAME GRID AND ACCELERATION CEILING: *no* WHERE ONE CARRIER IS EXCLUDED, *open* WHERE THE PAIR IS NOT SEPARABLE, *yes* FOR THE CONTROL ROW. A FITTED DRIFT IS REFINED INSIDE ONE GRID STEP AND MAY FALL BEYOND THE OUTERMOST TRIAL, HERE BY 0.72 OF A 59 Hz s^{-1} STEP, SO A VALUE ABOVE MAY EXCEED THE 12.0 nHz DEFAULT CEILING OF §4 ALTHOUGH THAT WINDOW'S OWN OUTERMOST TRIAL, 11.99 nHz , DOES NOT.

Star	blocks	Δt (d)	$\dot{\nu}_1$ (Hz s^{-1})	$\dot{\nu}_2$ (MHz)	$\Delta\nu$ miss one (σ) carrier?
61 Vir	X822/X82f	3.00+1092+3221+755.4			15 open
J1537–3319	X10359/X398	123–2000+2390+223.5			3.6 open
AU Mic	X1838/X319	310+1440+2645–159.0			— open
HD 14055	X2b2ee/X2c3bb	0.09–2390+4216+825.2			2447 no
	X24be0/Xd46	1.92+1134–3823+63.0			41 open
	X2c3bb/X3835	2.00+4216+1800–514.6			144 no
	X2b2ee/X3835	2.09–2390+1800+310.6			48 open
	X3835/Xd526	607+1800–2563–555.5			— open
	X2b2ee/Xd526	609–2390–2563–244.9			— open
	X3835/X25c5e	649+1800+2611+423.3			— open
	X2c3bb/X25c5e	651+4216+2611–91.3			— open
	X2b2ee/X25c5e	651–2390+2611+733.8			— open
β Pic	X3a90/Xa9df	1.02–1611–679+85.5			132 no
HD 285968	X108f/X5b4e	4.99–29–29–0.002			0.6 yes

Two sets of execution blocks enter no census statistic and calibrate instead (Table A1); both were searched with the frozen pipeline, both hold data on real targets and both are reported here as searches. The hold-out — 77 blocks, 315 windows, 36 stars and 214h on source, held out by a rule fixed before any of them was searched and sharing no execution block with the census (Appendix B.1) — contains 8 threshold crossings, the strongest at $T_\star = 8.64$, and not one of them outranks its own control positions: the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again: HD 285968 at $+9 \text{ km s}^{-1}$ from CO(2–1), whose 4 later blocks combine to $T_{\text{rep}} = +6.71$ at the carried drift, the value Table 7 prints, while the strongest single block reaches 5.87 at its own best drift. It is a molecular line, so its recovery is the test working in the positive direction here as well; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 7).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88h on source, with 8 crossings whose strongest is $T_\star = 12.7$ and none of which outranks its own controls. No systemic velocity is catalogued for any of them, so the set supports no limit and shows only that the pipeline behaves the same way on fields it was not tuned for. The third reserved set calibrates the spatial rank and is not external at all (Appendix B).

5.7. A stellar-frame multi-epoch stacked search

TABLE 6

THE TWO EVENTS, THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW; THE FIRST OF THEM IS FIG. 7. REPEAT EPOCHS MEASURED ARE THE BLOCKS AT WHICH THE PREDICTED STELLAR-FRAME CELL WAS READ, THE REMAINDER OF THEM COMING FROM THE ARCHIVAL EXTENSION. T_{pers} IS WHAT A CARRIER PERSISTING AT THE DISCOVERY AMPLITUDE S WOULD HAVE GIVEN THERE AND T_{rep} IS WHAT IS MEASURED, COMBINING EVERY REPEAT EPOCH BY INVERSE VARIANCE AT THE CARRIED DRIFT: ONE CELL AND ONE TRIAL, SO IT MAY BE NEGATIVE, WHILE THE TWO INDENTED READINGS BENEATH IT ARE MAXIMA AND SO POSITIVE ON AVERAGE WHATEVER IS THERE. THE NEAREST BLOCK ALONE FOLLOWS, BECAUSE THE COMBINATION REQUIRES THE CARRIER TO RETURN TO THAT CELL AT THE START OF EVERY BLOCK ACROSS THE WHOLE BASELINE AND THE SINGLE READING REQUIRES ONLY THE TWO OBSERVATIONS. THE EXCLUSION IS THE FRACTION OF S STILL RULED OUT AT THE TRIGGER, $(T_{\text{rep}} + 5)/T_{\text{pers}}$; THE DISPLACEMENT PERMITTED IS HALF A CHANNEL IN THE STAR'S FRAME. THE LAST ROW BUT ONE IS THE CHANCE $\Phi(5T_{\text{pers}}/T_\star - 5)$ THAT A CARRIER AT THE TRIGGER AMPLITUDE $5S/T_\star$ WOULD HAVE SATISFIED THE CRITERION, AND IN BRACKETS THE SAME AT S , WHICH IS SELECTED AS THE LARGEST OF MANY CELLS AND BIASED HIGH.

	61 Vir	CP–72 2713
Frequency, stellar frame (GHz)	346.323236	344.296455
as delivered, topocentric	346.313811	344.269746
Drift rate (Hz s^{-1})	+3182	–2056
$T_\star$	5.958	5.809
Highest control T	5.65	5.68
Fitted amplitude (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
of them in the primary census	3	1
census repeats with no	1	0
retained spectrum		
Nearest covering epoch, separation (h)	–1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, T_{pers}	9.70	6.30
T measured, matched drift, T_{rep}	+0.39	–0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Nearest block alone, T_{pers}	5.28	6.21
T_{rep}	–0.01	–0.57
Amplitude still excluded, $(T_{\text{rep}} + 5)/T_{\text{pers}}$	0.56 S	0.66 S
from the nearest block alone	0.95 S	0.71 S
Displacement permitted (km s^{-1})	± 0.211	± 0.213
a transmitter at 1 au would move	0.042	0.044
and over the longest baseline	1.53	1.08
Recovery chance at $5S/T_\star$ (at S)	1.00 (1.00)	0.66 (0.90)
Disposition	not confirmed	not confirmed

Combining the epochs of a star in that star's own rest frame deepens the limit. 55 of the 82 surveyed systems hold more than one searched execution block, and 58 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 315 stacked groups, 2177 epoch-windows and 1426 h, the flux limit improves by a median factor 1.52 over each star's best single epoch and 4.45 at most. 11 further stars stack too — 10 control stars beyond 40 pc and 1 covered only by archival blocks — and are excluded from those figures; over all 69 stacked stars, whose stacks are mostly shallow two-epoch ones, the median gain is 1.46. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are

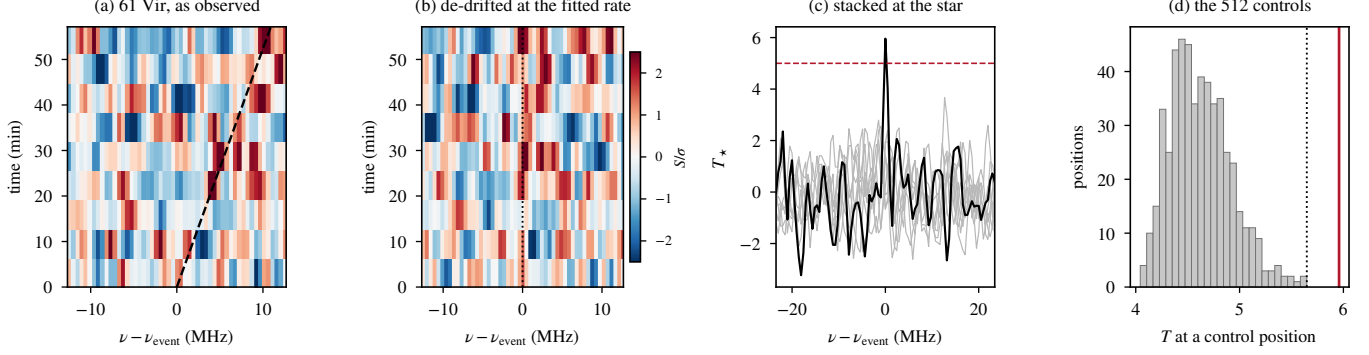


FIG. 7.— **The strongest crossing of the survey, and the control field it leads.** Frequencies are in the star’s rest frame. (a) The dynamic spectrum at 61 Vir’s position around the crossing at 346.3232 GHz, observed 2017 May 20, with the fitted drift of $+3182 \text{ Hz s}^{-1}$ as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin’s noise. At $119 \pm 20 \text{ mJy}$ against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them, and exists only in the coherent stack of all 38 minutes on source. (b) The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the observation. (c) The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. (d) The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted). The recurrence test, for this crossing and for the second one, CP-72 2713 at 344.2965 GHz , is in Table 6, which also gives the implied isotropic-equivalent powers, $5.1 \times 10^{14} \text{ W}$ at 8.5 pc here and $8.9 \times 10^{14} \text{ W}$ at 36.7 pc there.

TABLE 7

THE THRESHOLD CROSSINGS OF THE RESERVED HOLD-OUT, SEARCHED WITH THE SAME FROZEN PIPELINE AND DISPOSED OF BY THE SAME TESTS AS THE CENSUS. Δv_* IS THE OFFSET FROM THE NEAREST MASKED TRANSITION IN THE STAR’S OWN FRAME AND n_{rep} THE NUMBER OF COVERING REPEAT BLOCKS; T_{pers} AND T_{rep} ARE AS IN TABLE D1. THE ONE CROSSING THAT IS PRESENT AGAIN IS A CARBON MONOXIDE LINE.

Star	block	ν_{topo} (GHz)	T_* line	Δv_*	n_{rep}	T_{pers}	T_{rep}	disposition
HD 285968	Xdae5a_Xdd9b	230.5081	8.642 CO(2–1)	+8.7	4	11.6	+6.71	attributed; present again
HD 14055	X107376c_X6126	330.7123	5.513 $^{13}\text{CO}(3-2)$	+112.6	30	312.0	+0.56	unattributed; does not recur
TRAPPIST-1	Xd0e450_X259b	346.2454	5.326 SO(9 ₈ –8 ₇)	-306.6	5	11.1	-0.32	unattributed; does not recur
HD 172555	X107fa96_X1300f	344.6066	5.144 SO(8 ₈ –7 ₇)	+251.3	—	—	—	unattributed; no repeat coverage
HD 285968	Xdb4b9a_Xa7c	229.9129	5.106 CO(2–1)	-768.0	4	11.2	+0.21	unattributed; does not recur
HR 1010	Xc6b674_X23f2	346.2301	5.044 SO(9 ₈ –8 ₇)	-238.7	7	25.9	-0.80	unattributed; does not recur
HD 10647	Xf99e1_X9cdf	345.7059	5.017 SO(2 ₃ –2 ₁)	+37.1	13	142.1	-0.93	unattributed; does not recur
HD 170773	Xf294f_X1fb	345.5597	5.001 SO(2 ₃ –2 ₁)	-113.8	4	59.3	+1.45	unattributed; does not recur

added. The barycentric term alone swings by a median 8.2 km s^{-1} between the epochs of one star and by as much as 53.9 — many channels at the 488 kHz Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star’s catalogued identity rather than the directory the search ran in. Spectra with pathological archived visibility weights are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line: HD 285968’s molecular line collapses from a 19.26 -channel topocentric spread to 0.798 channels over 4 epochs spanning 44 d .

The criterion was fixed before any stellar position was examined: a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, measuring the whole test’s false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is $\sqrt{N_{\text{eff}}}$ and not $\sqrt{N}$, where $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median N_{eff}/N is 0.65 . What a stack realises is $\sqrt{N_{\text{eff}}}$ divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-

variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18 . Against $\sqrt{N}$ the realised gain is 0.78 , the shortfall being unequal epoch depth, so an epoch count alone misleads. HD 161868 stacks 13 epochs for a gain of only 1.15 : its N_{eff} is 1.7 , so $\sqrt{N_{\text{eff}}}$ is 1.29 , and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is *Proxima Centauri’s*, and it is a *Class B* limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr , all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census, 33 were searched afterwards from the same public archive and 8 come from the reserved hold-out, all observed over the census blocks’ own years. Stacking the census blocks alone, the set every other limit here uses, gives $\text{EIRP}_{90} = 2.2 \times 10^{13} \text{ W}$; adding the rest of the archive deepens that by 1.26 to $1.73 \times 10^{13} \text{ W}$, against $5.51 \times 10^{13} \text{ W}$ for the best single epoch on the same injection-calibrated convention. Its N_{eff} is only 7.9 , so $\sqrt{N_{\text{eff}}} = 2.80$ of the measured 3.19 is what combining the epochs buys, the rest being this spectrum’s own noise. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel, 20.9 km s^{-1} peak to peak, with no inter-epoch drift above $7.8 \times 10^{-2} \text{ Hz s}^{-1}$. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known

lines. They are β Pictoris' CO $J = 1-0$ at $+3.09 \text{ km s}^{-1}$ over 8 epochs ($Z = 45.3$) and its $J = 2-1$ at $+3.55 \text{ km s}^{-1}$ ($Z = 12.1$), agreeing to 0.46 km s^{-1} or 0.73 of a CO($1-0$) channel: the astrophysical positive control passing in the stack. The survey's own $\pm 50 \text{ km s}^{-1}$ mask leaves none of them unattributed. HD 285968 reaches $Z = 12.89$ on the line used above as the registration standard and is rejected by its own controls at 14.19: the emission is extended, as a cloud is and a transmitter is not.

Every stacked limit above holds only for a carrier that stays inside one channel. At CO($2-1$) one 488 kHz channel spans 0.63 km s^{-1} , the whole excursion a stacked carrier is allowed, peak to peak. A transmitter on a short-period giant planet, its line-of-sight velocity swinging by $\pm 30 \text{ km s}^{-1}$, crosses 94 of those channels peak to peak and de-registers completely, against 3.0 of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to $1.3 \times 10^{-1} \text{ Hz s}^{-1}$ over the median inter-epoch span of 44 d. No crossing of §5.6 drifts that slowly: the slowest measured is 519 Hz s^{-1} , which leaves one channel in 16 minutes, and in 33 of 38 cases the stack covering a crossing contains its discovery block. Toward the 20 stars carrying them the stack is a deeper limit on a non-drifting carrier, not a test of these events. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of §6.3 apply generally.

6. DISCUSSION

6.1. What is different about a millimetre technosignature search

The most transferable result of this experiment is not the non-detection but a measurement of what limits a carrier search at these wavelengths. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity. That is a statement about disposing of events and not about depth: the power a transmitter must radiate to be found here remains a median $1.6 \times 10^{15} \text{ W}$ per system.

Confusion, and not interference, sets the false-positive rate. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 56 threshold crossings, and no crossing in this survey is attributed to interference (Appendix E). 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, all but one the resolved disc of β Pictoris and the exception a cloud along the sightline to HD 48370. The remedy that works is a velocity-space mask evaluated in each star's own rest frame, and it is not free: it removes 6.4 per cent of the Class A union bandwidth.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its emission — foreground cloud, not a disc (Appendix C) — lies near a band edge, and that one window contributes

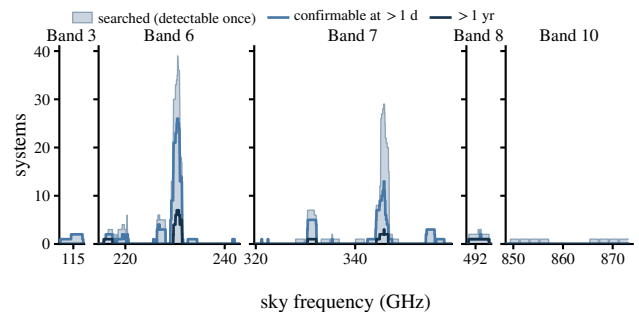


FIG. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is $\times 2.464$ the interior rate; without that window $\times 1.020$, and on a held-out half $\times 1.181$ ($p = 0.24$). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

The channelisation is inherited. 1249 of the 1651 windows have channels so coarse that every drift in the grid stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 402 of them. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude; set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the confirmation coverage $C_{\text{conf}}(\nu)$ be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency ν , so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.7 GHz of union bandwidth, 52 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 26 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

6.2. The search space this archive opens

The carrier search is 402 fine-channel windows toward 60 stellar systems, 47.7 GHz of union bandwidth; the 1249 coarse-channel windows are a supplementary spectral-excess experiment. Earlier searches above about 30 GHz reached nearby stars at one frequency only, and the one earlier millimetre search of comparable bandwidth worked in Band 3 toward far more distant stars (§2); Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J = 1-0$ control. Of the 10 MHz–115 GHz haystack axis of Wright et al. (2018)

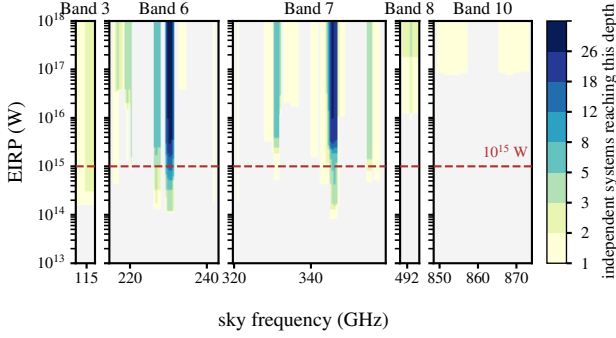


FIG. 9.— Where the experiment could have found something. Shading gives the number of independent systems in which an unresolved carrier of the plotted EIRP at the plotted sky frequency would have been recovered nine times in ten *in a single epoch*; unshaded means none. That is a recovery completeness and not a probability of confirmation: confirmation requires a second independent epoch covering the same cell, which Figure 8 maps and which no amount of transmitted power can supply. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is 1×10^{15} W.

this experiment covers 19.9 GHz, only 1.0 GHz of it carrier search because the Class A union begins at 114 GHz; the rest lies above that axis.

Union bandwidth overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 gives instead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems and 17 at the best-covered frequency near 231 GHz; with no limit on power that frequency reaches 39 of the 60 systems, and half the sample only above 3.4×10^{15} W.

The conventional measure for comparing programmes is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), $\text{CWTFM} = \zeta_{\text{AO}} \text{EIRP}_{\text{min}} / (N_{\star} \nu_{\text{rel}})$ with $\nu_{\text{rel}} = \Delta\nu / \nu_{\text{mid}}$, smaller being better. Those programmes quote EIRP_{min} at a bare detection threshold, and not at the same one: Enriquez et al. (2017) quote at $\text{S/N} = 25$ and Margot et al. (2023) at 10, where this search triggers at five (Table 1). The measure is linear in EIRP_{min} , so putting every programme on the common footing of five moves the published values — 0.85 becomes 0.17 and 0.14 becomes 0.072 — and it runs against this survey, making the comparison programmes deeper than their own papers report. Margot et al. (2023) publish no CWTFM; that value is the same expression on the three quantities they do give, 11 680 stars, 1.15–1.73 GHz and $\text{EIRP} > 1.35 \times 10^{13}$ W. This experiment takes no such factor, and on that common footing, with $N_{\star} = 60$ and $\nu_{\text{rel}} = 0.097$, the Class A experiment is 1.0×10^5 on the sample’s largest threshold power and 4.4×10^3 on the median system, a gap of 5.2×10^3 from the published Enriquez value that widens to 2.6×10^4 once that value is rescaled with the rest. On the EIRP_{90} adopted everywhere else here the same two figures of merit are 3.2×10^5 and 1.3×10^4 . Nearly all of the difference is the product $N_{\star} \nu_{\text{rel}}$: this is a far smaller haystack than a wideband centimetre survey of thousands of stars, and the measure compares surveys

rather than estimating occurrence rates.

6.3. What this experiment excludes, and what it does not

The signal these limits describe. Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while one of the archival blocks is on sky, and changing frequency linearly at no more than its own window’s ceiling, 12.0–13.3 nHz. Such a carrier would have produced a crossing nine times in ten above the EIRP_{90} of §5.1, and none survived toward any of the 60 systems the carrier search covers. Four conditions bound it as sharply as power does: the coverage is sparse and inherited (Fig. 9), the median system covered over only 1.84 GHz; the drift grid reaches an acceleration that contains every known planet of the sample but one, but not a transmitter on a rapidly rotating body (Fig. 3); emission broader than one channel is removed with the spectral baseline; and a carrier on for part of a block comes back at that fraction of its amplitude, while one absent throughout is not constrained at all.

Recurrence applies over a far smaller domain. A crossing is not a detection (Table 3), and where the archive observed only once no power makes one. Recurrence could be tested for 37 of the 60 systems at a frequency the first epoch covered (§5.6.1); for the other 23 the confirmation completeness is zero. Where it does apply, the test has power only against a carrier as strong as the trigger demands of the repeat, so a shallow repeat excludes nothing (§5.6).

The result is a non-detection and not an exclusion of persistent transmitters. An exclusion is read at the one cell the first epoch predicts, so the transmitter it tests is one that comes back to that cell, and four restrictions on that model are ours. A transmitter operating intermittently need not have been radiating during the second block. A second block need not cover the frequency at all, and over about half the searched bandwidth none does (§6.1). Orbital motion moves the carrier in the star’s own frame, and the interval the test permits is only half a channel, which an orbit at 1 au can carry a transmitter out of between the epochs (§5.6). A transmitter may also change frequency deliberately, and nothing here addresses that. Outside the adopted criterion, then, nothing is excluded: neither emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask at any strength, nor any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

An illustrative engineering benchmark. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D \nu / c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, a depth this survey reaches for 16 of the 60 Class A systems. The same gain makes its beam $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, 34 arcsec across and illuminating 1.7×10^{-9} of the sky. EIRP is not transmitter power — the same megawatt radiated isotropically would be 10^6 W — so these limits apply to beamed transmitters whose beam contains the Solar System.

Two blind spots. The search forms Stokes I as $(XX + YY)/2$, which retains the full intensity of a circularly polarised carrier; what is lost is not sensitivity but the dis-

crimination polarisation would offer against unpolarised astrophysical emission, because ALMA’s feeds are linear and the circularity lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s and not a choice; none of the “magic frequency” arguments that have guided centimetre-band searches falls in an ALMA band.

6.4. *What a dedicated ALMA technosignature experiment should do differently*

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere*: the 1249 coarse-channel windows cannot carry a carrier search at all. *Choose the targets and the frequencies*: an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 3.05 GHz of the 47.7 GHz Class A union that the mask costs, 6.4 per cent of it — all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning*: two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities*: rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. The two earlier searches to reach nearby stars above 30 GHz each covered a single line, over no more than 35 MHz, and neither used its repeats as a test of persistence (§2). What is new here is the bandwidth, at least 1363 times theirs, a grid of drift rates rather than a fixed frequency, and a test for recurrence; and an archive assembled for other science is at present the only way to reach that combination. The one ALMA band below those searched that holds any data on these stars is proprietary until 2027 and channelised far too coarsely to carry a carrier search.
2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of union bandwidth in 26 disjoint intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1249 windows toward 81 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform search of 404 public execution blocks, 4.1 TB of raw data, and between them they span 90–873 GHz.
3. EIRP₉₀ is a measured completeness of one epoch and not a threshold: the power at which nine of ten carriers injected into the real calibrated visibilities come back through the unchanged trigger and localisation at the star, inside a single execution block, for a carrier narrower than that window’s own channel; one spread over many channels meets a different limit entirely. The median system’s best window

reaches 1.6×10^{15} W, with a spread across systems of 8.7×10^{13} – 3.7×10^{16} W, and 24 systems reach 10^{15} W; carriers were injected into 49 of the 402 windows and the rest carry the class-average factor, whose measured dispersion moves that median by 2.2 per cent, so a single window’s value is an estimate. One bias runs one way only and is outside that calibration: an injected carrier suffers no atmospheric decorrelation where a real one would, which leaves the median limit optimistic by 1 per cent and the windows above 400 GHz by 8 per cent. No limit quoted here carries that correction, and for the 125 blocks whose phase stability cannot be measured it is conditional on their own calibration performance.

4. All 56 threshold crossings are reported with their dispositions, four of them in the one execution block that fails the data-quality criterion of Appendix D.2 and reported apart from the 52 in quality-passing data. Of the 56, 21 fall within $\pm 50 \text{ km s}^{-1}$ of a catalogued transition at the star’s own velocity, a coincidence that is reproducible and involves no judgement. Of those, three are with a species no debris disc has been shown to carry and are reported rather than acted on, leaving 18 attributed and 38 not. The mask costs 6.4 per cent of the Class A union bandwidth. Recording every cell above the trigger, rather than each window’s strongest, adds 21 more: three survive attribution and none recurs, so no disposition changes. Chance, measured from each window’s own control positions and channel grid, predicts 28.9 unattributed Class A crossings over the census against the 32 found there, or 30.8 with the measured star-to-control transfer applied, and resampling the execution blocks allows 12–48. The count is compatible with that empirical control expectation, which is a weaker statement than a demonstration that the crossings are noise: the width of the range is its limit, and an added population of as many as 19 crossings would sit inside it unremarked. The one instrumental mechanism that could raise the statistic at the star and not at its controls is a calibration residual varying with frequency, and against the continuum the pipeline records at the extraction position — dust inside the beam included — nine windows toward two stars would admit one at 3.8–13 per cent, inside the documented range. In every one of them the largest statistic at any channel and drift is 3.1 and none holds a crossing, which puts such a residual below 6.6 per cent: the mechanism is available to those windows and is measured not to be acting in them.
5. Every unattributed crossing was tested for recurrence in the star’s own frame, over the whole of each further window covering the event frequency and at every drift searched, because an accelerating carrier need not return to one channel. None was found again: the largest statistic at any predicted cell is 4.71 against $T_{\text{trig}} = 5$. The same criterion recovers 13 of the 15 line-attributed crossings tested, which is the positive control of the stage. Of the 33 crossings that carry an exclusion, the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude with individual limits from 0.017 to 1.00, so the median is

not a bound on the set; combining every covering block assumes a carrier that returns to one stellar-frame channel at the start of each, and where the repeat is shallow the criterion has little power and its silence is not an exclusion. Judged at the trigger amplitude rather than at the discovery amplitude, 7 of these crossings are untested rather than excluded, among them one of the two that lead their own control fields. What these tests exclude is a carrier that holds its stellar-frame frequency to within $\pm 0.08\text{--}\pm 0.63\text{ km s}^{-1}$ of the first epoch's and radiates through both observations — an interval orbital motion alone can carry a transmitter out of (§5.6). The conclusion this supports is that no unresolved spectral carrier satisfying the adopted recurrence criterion was confirmed, and not that persistent transmitters are excluded above a power threshold.

6. What this survey could search and what it could confirm are different extents. Of the 60 systems the carrier search covered, 37 were observed again more than 1 day later at a frequency the first block covered; for the other 23 confirmation was beyond reach at any sensitivity, and about those systems the null reported here says nothing at all.
7. Combining each surveyed star's epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri the stacked limit, measured by injection into the stack, reaches $2.2 \times 10^{13}\text{ W}$ over the primary census alone and $1.73 \times 10^{13}\text{ W}$ once the rest of the archive is added, in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to within one channel across 6.3 yr.
8. These results constrain unresolved carriers only within the domain actually searched (§6.3). The drift reach is part of that domain: the grid stops at 12.0 nHz by default, 6 per cent of the range recommended for a general carrier search (Sheikh et al. 2019), so an unbound transmitter, one on a rapidly rotating body, or one whose combined orbital and rotational acceleration exceeds that ceiling is not constrained here at all. What transfers beyond the domain is the obstacle, that channelisation chosen by other observers, and not collecting area, decides which kind of search is possible at all.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection-recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table D1.

FUNDING

This work received no dedicated funding. It used public archival data and no observing time was awarded for it.

COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 July 11 and belong to 65 ALMA proposals, listed in Table A3, whose principal investigators and teams we thank. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France), of the NASA Exoplanet Archive (Akeson et al. 2013), and of the Splatalogue spectral-line database (Remijan et al. 2007).

DATA AVAILABILITY

Every observation this paper rests on is public. The 404 primary-census execution blocks and the blocks of every other set named in Table A1 are held in the ALMA Science Archive and are retrieved there by the execution-block identifier, which Tables D1 and D2 give for every one of the 56 threshold crossings. The ancillary catalogues are public in the same sense: *Gaia* DR3 for the astrometry and the colours, the NASA Exoplanet Archive for the planet counts, and the CDMS and JPL line catalogues through Splatalogue for the mask, queried once on 2026-10-06.

The derived results are in the paper. Table A1 partitions the blocks, windows, stars, systems and hours of every set the search defines; Table A2 carries every headline count, the five fates of the unprocessed blocks and the crossing partition; Tables D1 and D2 list all 56 crossings with their frequencies in both frames, the stellar and control statistics, the nearest masked transition and the recurrence test; Table 6 gives the two leading events in full; and Table 7 does the same for the reserved hold-out. The method is set out in enough detail to be reimplemented, and no separate data deposit accompanies the paper.

Three choices that could create or remove an unattributed crossing carry an identifier anyone can check. The calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were committed to the project repository as c75069040eab52ec1336376a4dae62114bb281a3 on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The molecular-line list is a single dated query of a public line catalogue (2026-10-06), so it is specified rather than described, and it post-dates the fixing of the disposition criteria on 2026 September

11 (f5f52e7804f1baad99515ea2d46ddd7e4e4cd546), as does every species in it, so no species in the list was recognised late or early with respect to them.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

SUPPORTING INFORMATION

The following appendices are published as online Supporting Information with the electronic version of this paper. They are referred to by letter throughout the text, and their sections, figures and tables keep the numbering used here.

- **Appendix A.** Spectral response, frequency conventions and the primary-beam response: the channel response of the correlator and what it does to an unresolved carrier, the three frequency frames and the

transformations between them, the intra-integration smearing correction applied to every limit, and the primary-beam attenuation that turns a measured noise into a flux at the star.

- **Appendix B.** The spatial rank and what it can support: what the comparison with the 512 control positions does and does not establish, the reserved out-of-sample hold-out that calibrates it, the channel-dependent calibration mechanisms and the measurement that excludes them, and the null of the recurrence statistic.
- **Appendix C.** The line mask: the velocity criterion, its evaluation in the star’s own rest frame with the local standard of rest as a check, the published carbon-monoxide surveys each sightline is checked against, the masked bandwidth and what it costs, and the one species treated as a note rather than as an attribution.
- **Appendix D.** The false-alarm expectation and the control ensemble: what the expectation is conditioned on, the execution-block quality criterion, the calibration and the resolution of the control ensemble, the trials budget, and the crossing ledger itself.
- **Appendix E.** Radio-frequency interference screening.

REFERENCES

- Akeson R. L., et al., 2013, *PASP*, 125, 989
 Astropy Collab., 2022, *ApJ*, 935, 167
 Backus P. R., Project Phoenix Team, 2002, in Stanimirović S., Altschuler D., Goldsmith P., Salter C., eds, *ASP Conf. Ser. Vol. 278, Single-Dish Radio Astronomy: Techniques and Applications*. Astron. Soc. Pac., San Francisco, p. 525
 Burton K., MacGregor M. A., Osten R. A., 2022, *ApJL*, 939, L6
 CASA Team, 2022, *PASP*, 134, 114501
 Cataldi G., et al., 2023, *ApJ*, 951, 111
 Choza C., et al., 2024, *AJ*, 167, 10
 Cocconi G., Morrison P., 1959, *Nature*, 184, 844
 Cortes P. C., et al., 2026, *ALMA Cycle 13 Technical Handbook*, ALMA Doc. 13.3, ver. 1.0, doi:10.5281/zenodo.18793803
 Czech D. J., et al., 2021, *PASP*, 133, 064502
 Dame T. M., Hartmann D., Thaddeus P., 2001, *ApJ*, 547, 792
 Dame T. M., Thaddeus P., 2022, *ApJS*, 262, 5
 Dent W. R. F., et al., 2014, *Science*, 343, 1490
 Drake F. D., 1961, *Physics Today*, 14(4), 40
 Enriquez J. E., et al., 2017, *ApJ*, 849, 104
 Gaia Collab., Vallenari A., et al., 2023, *A&A*, 674, A1
 Garrett M. A., 2026, *MNRAS*, in press (arXiv:2605.10212)
 Ginsburg A., et al., 2019, *AJ*, 157, 98
 Harris C. R., et al., 2020, *Nature*, 585, 357
 Hunter T. R., Phillips N. M., Brogière D., González J., 2018, *Improving the Calibration of Atmospheric Spectral Features in ALMA Data*. NA ALMA 2016 Development Study Final Report, 2018 March 2
 International Telecommunication Union, 2024, *Radio Regulations*, Edition of 2024. ITU, Geneva
 Jacobson-Bell B., et al., 2025, *AJ*, 169, 206
 Johnson O. A., et al., 2023, *AJ*, 166, 193
 Kamenó S., 2014, *ALMA Cross Correlation Bandpass Stability*. ALMA Technical Note 15 (draft), 2014 December 17
 Li J.-K., Tao Z.-Z., Zhang T.-J., 2026, *ApJ*, 1006, 163
 Ma P. X., et al., 2023, *NatAs*, 7, 492
 MacGregor M. A., et al., 2020, *ApJ*, 891, 80
 Manunza L., et al., 2025, *AcAau*, 233, 155
 Margot J.-L., et al., 2023, *AJ*, 166, 206
 Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2025, *MNRAS*, 536, 2127
 Matrà L., et al., 2017, *MNRAS*, 464, 1415
 Mauersberger R., Wilson T. L., Rood R. T., Bania T. M., Hein H., Linhart A., 1996, *A&A*, 306, 141
 Morrison I. S., et al., 2023, *PASA*, 40, e019
 Müller H. S. P., et al., 2005, *JMoSt*, 742, 215
 Pardo S., et al., 2025, *AJ*, 170, 12
 Pickett H. M., et al., 1998, *JQSRT*, 60, 883
 Planck Collab., Adam R., et al., 2016, *A&A*, 594, A10
 Price D. C., et al., 2020, *AJ*, 159, 86
 Privon G., et al., 2026, *ALMA Cycle 13 Proposer’s Guide*, ALMA Doc. 13.2, ver. 1.0, doi:10.5281/zenodo.18793859
 Remijan A. J., Markwick-Kemper A., ALMA Working Group on Spectral Line Frequencies, 2007, in *American Astronomical Society Meeting Abstracts* 211, 132.11
 Richards A. M. S., et al., 2022, *Self-calibration and improving image fidelity for ALMA and other radio interferometers*, ALMA Memo 620
 Sheikh S. Z., Wright J. T., Siemion A. P. V., Enriquez J. E., 2019, *ApJ*, 884, 14
 Sheikh S. Z., 2020, *IJAsB*, 19, 237
 Sheikh S. Z., et al., 2021, *Nature Astronomy*, 5, 1153
 Sheikh S. Z., et al., 2025, *AJ*, 169, 118
 Steffes P. G., DeBoer D. R., 1994, *Icarus*, 107, 215
 Stephens G. L., et al., 2002, *Bull. Am. Meteorol. Soc.*, 83, 1771
 Tarter J., 2001, *ARA&A*, 39, 511
 Tremblay C. D., Tingay S. J., 2024, *ApJ*, 972, 76
 Vidal C., et al., 2026, arXiv:2605.21093 (preprint)
 Virtanen P., et al., 2020, *NatMe*, 17, 261
 Wehr T., et al., 2023, *Atmos. Meas. Tech.*, 16, 3581
 Wenger M., et al., 2000, *A&AS*, 143, 9
 Wright J. T., et al., 2018, *AJ*, 156, 260
 Zhang Z.-S., et al., 2020, *ApJ*, 891, 174